

5. Rozwiąż równania rekurencyjne:

a) $x_n = 2x_{n-1} + x_{n-2} - 2x_{n-3}$, $x_0 = 0$, $x_1 = 2$, $x_2 = 0$

b) $x_n = x_{n-1} + x_{n-2}$, $x_0 = 0$, $x_1 = 1$

c) $x_{n+2} - 5x_{n+1} + 6x_n = 2^n$, $x_0 = 0$, $x_1 = 1$

d) $x_{n+2} - 5x_{n+1} + 6x_n = 5^n + n$, $x_0 = 0$, $x_1 = 1$

e) $x_{n+1} = x_n + n^3$, $x_0 = 1$,

a) $x_n - 2x_{n-1} - x_{n-2} + 2x_{n-3} = 0$

$$r^3 - 2r^2 - r + 2 = 0$$

$$r^2(r-2) - 1(r-2) = 0$$

$$(r^2 - 1)(r - 2) = 0$$

$$r_1 = 1, r_2 = -1, r_3 = 2$$

$$x_n = c_1 1^n + c_2 (-1)^n + c_3 2^n$$

$$\begin{cases} 0 = c_1 + c_2 + c_3 \\ 2 = c_1 - c_2 + 2c_3 \\ 0 = c_1 + c_2 + 4c_3 \end{cases}$$

$$c_3 = 0, c_2 = -1, c_1 = 1$$

$$x_n = 1 - (-1)^n$$

5. Rozwiąż równania rekurencyjne:

a) $x_n = 2x_{n-1} + x_{n-2} - 2x_{n-3}$, $x_0 = 0$, $x_1 = 2$, $x_2 = 0$

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c) $x_{n+2} - 5x_{n+1} + 6x_n = 2^n$, $x_0 = 0$, $x_1 = 1$

d) $x_{n+2} - 5x_{n+1} + 6x_n = 5^n + n$, $x_0 = 0$, $x_1 = 1$

e) $x_{n+1} = x_n + n^3$, $x_0 = 1$,

b) $X_N - X_{N-1} - X_{N-2}$

$$R^2 - R - 1 = 0$$

$$R_{1,2} = \frac{1 \pm \sqrt{5}}{2}$$

$$X_N = C_1 \left(\frac{1 + \sqrt{5}}{2} \right)^N + C_2 \left(\frac{1 - \sqrt{5}}{2} \right)^N$$

$$\begin{cases} 0 = C_1 + C_2 \\ 1 = C_1 \frac{1 + \sqrt{5}}{2} + C_2 \frac{1 - \sqrt{5}}{2} \end{cases}$$

$$1 = C_1 \frac{1 + \sqrt{5}}{2} - C_1 \frac{1 - \sqrt{5}}{2}$$

$$1 = C_1 \left(\frac{1 + \sqrt{5}}{2} - \frac{1 - \sqrt{5}}{2} \right)$$

$$C_1 = \frac{1}{\sqrt{5}} \quad C_2 = -\frac{1}{\sqrt{5}}$$

$$X_N = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^N - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^N$$

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e) $x_{n+1} = x_n + n^3$, $x_0 = 1$,

c)

1) $R^2 - 5R + 6 = 0$

Wyliczamy równanie jednorodne

$(R-2)(R-3) = 0$

$R_1 = 2, R_2 = 3$

$X_N = C_1 2^N + C_2 3^N$

2) $X_N^p = C(N) \cdot 2^N$

$X_N^p = C \cdot 2^N$

2 jest rozwiązaniem równania jednorodnego więc musimy pomnożyć razy n

$(C(N+2)2^{N+2}) - 5(C(N+1)2^{N+1}) + 6(CN2^N) = 2^N \quad / : 2^N$

$C(N+2) \cdot 4 - 5 \cdot C(N+1) \cdot 2 + 6CN = 1$

$4CN + 8C - 10C(N+1) + 6CN = 1$

$10CN + 8C - 10CN - 10C = 1$

$C = -\frac{1}{2}$

$X_N^p = -\frac{1}{2} \cdot N \cdot 2^N$

3) $X_N = C_1 \cdot 3^N + C_2 \cdot 2^N - \frac{1}{2} \cdot N \cdot 2^N$

$\begin{cases} 0 = C_1 + C_2 \end{cases}$

$\begin{cases} 1 = 3C_1 + 2C_2 - 1 \end{cases}$

$C_1 = 2 \quad C_2 = -2$

$X_N = 2 \cdot 3^N - 2 \cdot 2^N - \frac{1}{2} \cdot N \cdot 2^N$